

An Analysis of DALL-E image generation

DALL-E image generation

DALL-E image generation -- Part 1

Ramesh, Aditya; Pavlov, Mikhail; Goh, Gabriel; Gray, Scott; Voss, Chelsea; Radford, Alec; Chen, Mark; Sutskever, Ilya (26 February 2021). "Zero-Shot Text-to-Image Generation". arXiv:2102.12092 [cs.CV].. The original report on DALL-E. DALL-E 3 System Card DALL-E 3 paper by OpenAI DALL-E 2 website Craiyon website

DALL-E image generation -- Part 2

DALL-E, DALL-E 2, and DALL-E 3 (stylised DALL·E) are text-to-image models developed by OpenAI using deep learning methodologies to generate digital images from natural language descriptions known as prompts. The first version of DALL-E was announced in January 2021. In the following year, its successor DALL-E 2 was released. DALL-E 3 was released natively into ChatGPT for ChatGPT Plus and ChatGPT Enterprise customers in October 2023, with availability via OpenAI's API and "Labs" platform provided in early November. Microsoft implemented the model in Bing's Image Creator tool and plans to implement it into their Designer app. With Bing's Image Creator tool, Microsoft Copilot runs on DALL-E 3. In March 2025, DALL-E-3 was replaced in ChatGPT by GPT Image's native image-generation capabilities.

DALL-E image generation -- Part 3

Open-source implementations Since OpenAI has not released source code or weights for any of the three models, there have been several attempts to create open-source models offering similar capabilities. Released in 2022 on Hugging Face's Spaces platform, Craiyon (formerly DALL-E Mini until a name change was requested by OpenAI in June 2022) is an AI model based on the original DALL-E that was trained on unfiltered data from the Internet. It attracted substantial media attention in mid-2022, after its release due to its capacity for producing humorous imagery. Another example of an open source text-to-image model is Stable Diffusion by Stability AI.

DALL-E image generation -- Part 4

Technical limitations DALL-E 2's language understanding has limits. It is sometimes unable to distinguish "A yellow book and a red vase" from "A red book and a yellow vase" or "A panda making latte art" from "Latte art of a panda". It generates images of an astronaut riding a horse when presented with the prompt "a horse riding an astronaut". It also fails to generate the correct images in a variety of circumstances. Requesting more than three objects, negation, numbers, and connected sentences may result in mistakes, and object features may appear on the wrong object.

Additional limitations include generating text, ambigrams and other forms of typography, which often results in dream-like gibberish. The model also has a limited capacity to address scientific information, such as astronomy or medical imagery.

DALL-E image generation -- Part 5

DALL-E 2 DALL-E 2 uses 3.5 billion parameters, a smaller number than its predecessor. Instead of an autoregressive Transformer, DALL-E 2 uses a diffusion model conditioned on CLIP image embeddings, which, during inference, are generated from CLIP text embeddings by a prior model. This is the same architecture as that of Stable Diffusion, released a few months later.

DALL-E image generation -- Part 6

Given an existing image, DALL-E 2 and DALL-E 3 can produce "variations" of the image as individual outputs based on the original, as well as edit the image to modify or expand upon it. The "inpainting" and "outpainting" abilities of these models use context from an image to fill in missing areas using a medium consistent with the original, following a given prompt. For example, this can be used to insert a new subject into an image, or expand an image beyond its original borders. According to OpenAI, "Outpainting takes into account the image's existing visual elements — including shadows, reflections, and textures — to maintain the context of the original image."

DALL-E image generation -- Part 7

DALL-E 3 While a technical report was written for DALL-E 3, it does not include training or implementation details of the model, instead focusing on the improved prompt following capabilities developed for DALL-E 3.

DALL-E image generation -- Part 8

Capabilities DALL-E can generate imagery in multiple styles, including photorealistic imagery, paintings, and emoji. It can "manipulate and rearrange" objects in its images, and can correctly place design elements in novel compositions without explicit instruction. Thom Dunn writing for BoingBoing remarked that "For example, when asked to draw a daikon radish blowing its nose, sipping a latte, or riding a unicycle, DALL-E often draws the handkerchief, hands, and feet in plausible locations." DALL-E showed the ability to "fill in the blanks" to infer appropriate details without specific prompts, such as adding Christmas imagery to prompts commonly associated with the celebration, and appropriately placed shadows to images that did not mention them. Furthermore, DALL-E exhibits a broad

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understanding of visual and design trends. DALL-E can produce images for a wide variety of arbitrary descriptions from various viewpoints with only rare failures. Mark Riedl, an associate professor at the Georgia Tech School of Interactive Computing, found that DALL-E could blend concepts (described as a key element of human creativity). Its visual reasoning ability is sufficient to solve Raven's Matrices (visual tests often administered to humans to measure intelligence).

DALL-E image generation -- Part 9

DALL-E DALL-E has three components: a discrete VAE, an autoregressive decoder-only Transformer model (12 billion parameters) similar to GPT-3, and a CLIP pair of image encoder and text encoder. The discrete VAE can convert an image to a sequence of tokens, and conversely, convert a sequence of tokens back to an image. This is necessary as the Transformer model does not directly process image data. The input to the Transformer model is a sequence of tokenised image caption followed by tokenised image patches. The image caption is in English, tokenised by byte pair encoding (vocabulary size 16384), and can be up to 256 tokens long. Each image is a 256×256 RGB image, divided into 32×32 patches of 4×4 each. Each patch is then converted by a discrete variational autoencoder to a token (vocabulary size 8192). DALL-E was developed and announced to the public in conjunction with CLIP (Contrastive Language-Image Pre-training). CLIP is a separate model based on contrastive learning that was trained on 400 million pairs of images with text captions scraped from the Internet. Its role is to "understand and rank" DALL-E's output by predicting which caption from a list of 32,768 captions randomly selected from the dataset (of which one was the correct answer) is most appropriate for an image. A trained CLIP pair is used to filter a larger initial list of images generated by DALL-E to select the image that is closest to the text prompt.